

Corrections précises — après quelle ligne + quoi écrire

Découverte : ta §4.2.3 (p.20) dit déjà « orthogonality is strictly load-bearing », mais ta §3.1.3 (p.9) dit « exclusively training-time ... zero overhead ». **C'est la contradiction que le Reviewer 3.2 a vue.** Le fix ci-dessous rend §3.1.3 cohérent. Surligné jaune = texte à ajouter.

1.2 / 1.3 / 3.1 — Fuzzy

§3.1.1, p.8 — APRÈS la ligne 270 (après la phrase « ...acts as a prior distribution over the emotion classes. »)

INSÈRE ce paragraphe surligné :

Both RAF-DB and FER2013 provide a single label per image; we use no multi-annotator metadata. For a single-label corpus $n_c = \mathbb{1}[c=y]$, so Eq.(2) reduces to $\mu_c = (\mathbb{1}[c=y] + \alpha \cdot \pi_c) / (1 + \alpha)$, generated algorithmically, not from annotator votes. We set $\alpha = 0.1$ (the standard label-smoothing value) and take π as the empirical class-frequency prior; softening the one-hot target reflects the intrinsic ambiguity of facial expressions and improves calibration.

3.2 — Gram-Schmidt (LE point critique)

§3.1.3, p.9 — REMPLACE les lignes 319-323 (la phrase « ...operate exclusively as training-time regularization mechanisms... ensuring zero computational overhead... »)

REPLACE par ce texte surligné (rend §3.1.3 cohérent avec §4.2.3) :

As shown in Fig. 2, the fuzzy-label calculation is training-only. Gram-Schmidt, however, is load-bearing at inference: naively removing it collapses RAF-DB accuracy ($0.858 \rightarrow 0.073$). Deployment is therefore a two-step process — a fold-out rewrite removes the DPU-hostile operator, and a short post-fold fine-tuning (10 epochs, no orthogonality) lets the remaining convolutions re-absorb its function, recovering to 0.849. The benefit is internalized into the weights by fine-tuning, not obtained for free.

3.4 — Disgust / ambiguïté

§4.2.2, p.17 — APRÈS la ligne 567 (après « ...focused strictly on regions of high visual ambiguity. »)

AJOUTE cette phrase surlignée (complète Fig.14/15 déjà présentes) :

In a matched-backbone comparison (both YOLOv8n), FLEO improves the hardest minority class by the largest margin — disgust: +4.5 AP ($0.510 \rightarrow 0.554$) — while keeping overall mAP@0.5 at parity (0.817 vs 0.812), confirming that the gains concentrate on the most ambiguous, low-support class.

3.4 — Control (baseline +10 ep)

§4.2.3, p.20 — APRÈS la ligne 579 (après « ...fold-out penalty of only $\Delta_{\text{fold}} = 0.009...$ »)

AJOUTE cette phrase surlignée :

As a control for the fine-tuning effect, a converged baseline YOLOv8n trained for the same 10 additional epochs under the identical schedule improves by only +0.2 mAP@0.5 ($0.812 \rightarrow$

0.814); the folded-graph recovery is therefore genuine re-adaptation, not a generic benefit of extra optimization.

1.1 — Ablation de d (avec la TABLE)

§4.2.3, p.20 — APRÈS le paragraphe de la Table 7 (après la phrase PTQ, ~ligne 583)

AJOUTE la phrase surlignée + insère la Table (code LaTeX plus bas) :

We further ablate the per-emotion subspace width $d \in \{4, 8, 16, 32\}$ on RAF-DB (Table 8): accuracy remains within a 3-point band with no monotonic trend, while the FLEO cost grows $0.50/1.0/2.05/4.30\times$. We therefore adopt $d = 8$ as the accuracy-cost balance.

2.9 — YOLO11 / YOLO26

§4.4, p.26 — dans la Comparative Study

AJOUTE cette phrase surlignée :

We additionally evaluate YOLO11-FLEO ($\text{mAP}@0.5 = 0.802$). YOLO26 could not be integrated (its detection head uses an incompatible loss-output format). YOLOv8n remains preferred: it is convolution-only and thus DPU-native, whereas YOLO11's attention (C2PSA) blocks are not compilable as a single DPU subgraph.

Code LaTeX de la Table d'ablation (Point 1.1)

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\begin{table}[h]\centering
\caption{Subspace-width ablation on RAF-DB (YOLOv8n).}
\begin{tabular}{ccccc}
\toprule
$d$ & channels & $7d$ & mAP@0.5 & mAP@0.5:0.95 & FLEO cost \\
\midrule
4 & 28 & 0.807 & 0.801 & $0.50\times$ \\
\textbf{8} & \textbf{56} & \textbf{0.790} & \textbf{0.759} & $\mathbf{1.00\times}$ \\
16 & 112 & 0.806 & 0.774 & $2.05\times$ \\
32 & 224 & 0.820 & 0.813 & $4.30\times$ \\
\bottomrule
\end{tabular}
\end{table}
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Cosmétiques (titre, orthographe l.85-87, suppression §l.99-103, affiliations, figures) : voir **Complete_Edit_Guide.pdf**.